

Colton Kucsan

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 Wind Gap, PA

Skills

- Experienced in DFMEA and risk analysis in support of design control requirements
- Strong analytical and problem-solving skills applied to complex engineering challenges
- Strong mechanical aptitude and hands-on manufacturing experience
- Product design ownership
- Proficient in SolidWorks, Pro/E Creo, and AutoCAD
- ISO 13485 Lead Auditor Certified
- Effective communication and cross-functional collaboration

Education

Master of Science, Mechanical Engineering
Purdue University

Bachelor of Science, Mechanical Engineering
University of Pittsburgh

Experience

Hovertech International - Allentown, PA

Product Development Engineer II

January 2026 – Present

- Lead competitive product evaluations, including testing and research strategy, to benchmark performance in key product characteristics and translate findings into actionable design recommendations
- Develop and own test protocols, and lead design validation and verification activities, ensuring compliance with product and regulatory requirements
- Lead and oversee design changes to existing products, evaluating impacts to performance, cost, manufacturability, and regulatory compliance while coordinating cross-functional approval and implementation
- Lead design inputs, DFMEA, and risk analysis activities to support design control requirements, identifying and mitigating potential failure modes throughout the product development lifecycle
- Lead material sourcing and investigation efforts, evaluating trade-offs to meet design, performance, and cost requirements, and providing technical guidance on material selection
- Manage relationships with external manufacturers on design, manufacturing, and quality issues, serving as the primary technical point of contact and driving resolution of complex issues
- Drive the development of manufacturing methods, including new tooling and equipment, partnering with manufacturing and quality teams to optimize for cost, quality, and scalability
- Conduct supplier audits as a certified Auditor to ISO 13485 standards, assessing quality management system compliance and driving corrective action to mitigate supply chain risk
- Utilize 3D printing (FDM/SLA) to rapidly prototype parts and design concepts for early-stage validation and stakeholder review, fabricate custom fixtures to support test protocol execution, and produce functional prototypes used to inform and de-risk downstream metal manufacturing.

Product Development Engineer I

March 2023 – January 2026

- Evaluated competitive products through structured testing and research to benchmark performance across key product characteristics
- Established test protocols and executed design validation and verification activities to confirm compliance with product requirements
- Collaborated with quality and suppliers to develop manufacturing methods, including new tooling and equipment, supporting production readiness
- Conducted material sourcing and investigation to identify solutions meeting design, performance, and product requirements
- Partnered with external manufacturers to resolve design, manufacturing, and quality issues throughout the product lifecycle
- Developed foundational 3D printing skills (FDM/SLA), applying design-for-additive geometry and material considerations to prototype parts

LaFrance Corp - Concordville, PA

Project Manager

February 2020 – March 2023

- Collaborated with engineering teams and account managers to identify and fulfill account-specific requirements across a diverse project portfolio
- Conducted design reviews to assess and ensure manufacturing feasibility prior to production
- Engineered part-specific process flows to inform cost estimates for account teams and provide clear manufacturing instructions to production teams
- Managed multiple concurrent projects, prioritizing effectively to meet competing deadlines
- Oversaw production operations in China, ensuring project timelines were met and final products aligned with customer specifications

SMS Group - Pittsburgh, PA

Engineering Co-op

September 2017 – July 2019

- Produced design proposals for components in accordance with company standards and specifications
- Performed FEA analysis using Creo to determine component properties, including weight, center of mass, and second moments of inertia
- Conducted stress and fatigue analyses on machine components to evaluate performance and durability
- Created and assembled three-dimensional digital models from technical drawings, and generated technical drawings from digital models

Butch's Rod and Resto - Hellertown, PA

Auto Restoration Technician

June 2014 – August 2017

- Designed and fabricated sheet metal components using metal forming and shaping techniques to meet project-specific specifications
- Fabricated custom automotive components tailored to the unique requirements of each restoration
- Gained hands-on technical experience repairing and restoring vehicles across a range of makes, models, and conditions
- Managed competing deadlines across multiple concurrent projects, including working overtime as needed to meet project timelines
- Coordinated and executed vehicle transport and delivery to customers, often over long distances, ensuring timely and satisfactory completion of projects